Milestone 6 UMLs

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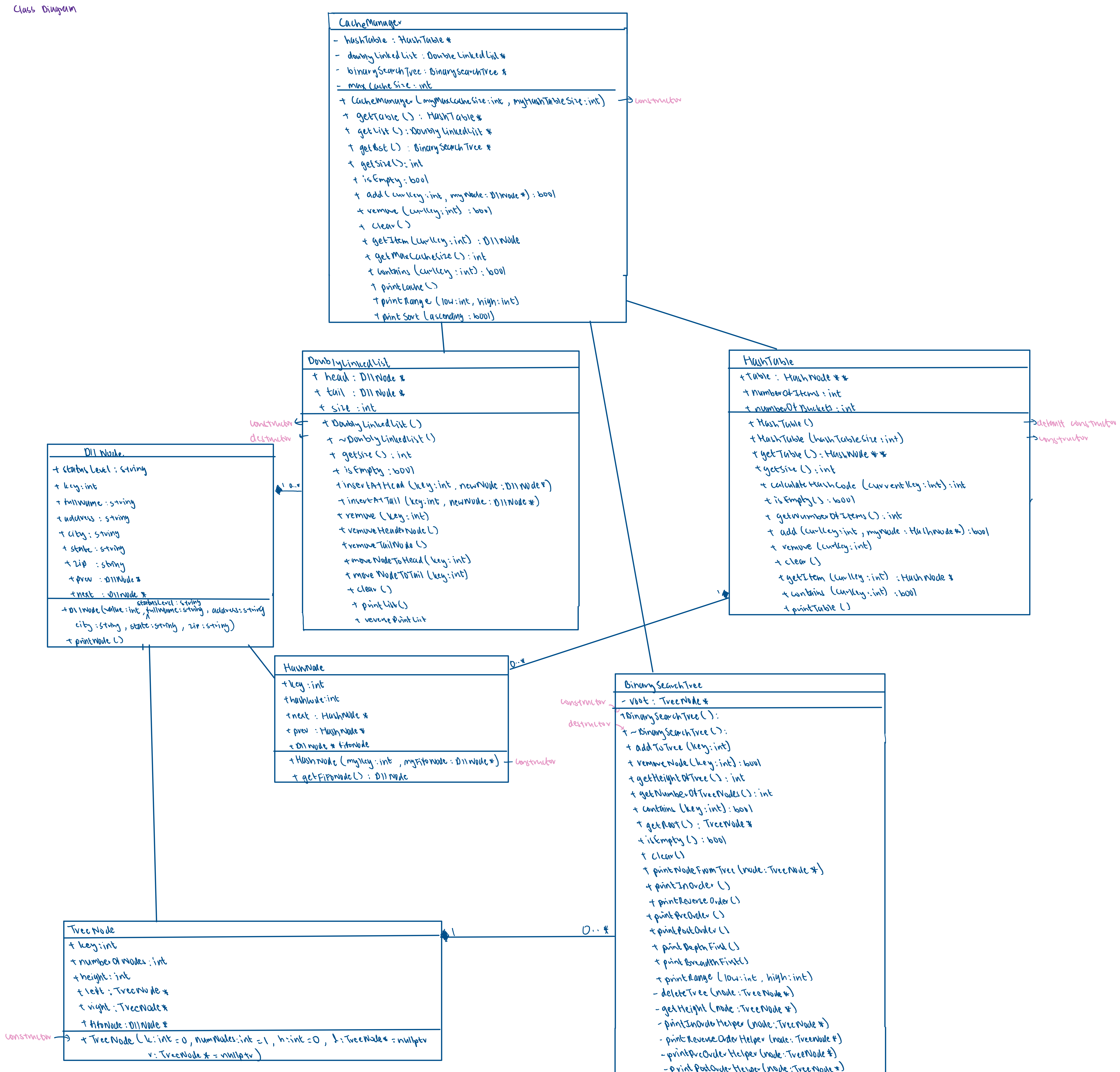
Class Diagram


CacheManager
- hashTable : HashTable *
- doublyLinkedList : DoubleLinkedList *
- binarySearchTree : BinarySearchTree *
- maxCacheSize : int
+ CacheManager (myMaxCacheSize : int , myHashTableSize : int)
constructor
+ getTable () : HashTable *
+ getList () : DoublyLinkedList *
+ getBst () : BinarySearchTree *
+ getSize () : int
+ isEmpty : bool
+ add (currKey : int , myNode : DllNode *) : bool
+ remove (currKey : int) : bool
+ clear ()
+ getItem (currKey : int) : DllNode
+ getMaxCacheSize () : int
+ contains (currKey : int) : bool
+ printCache ()
+ printRange (low : int, high : int)
+ printSort (ascending : bool)
DoublyLinkedList
+ head : DllNode *
+ tail : Dll Node *
+ size : int
constructor
destructor
+ DoublyLinkedList ()
+ ~DoublyLinkedList ()
+ getSize () : int
+ isEmpty : bool
+ insertAtHead (key : int , newNode : DllNode *)
+ insertAtTail (key : int , newNode : DllNode *)
+ remove (key : int)
+ removeHeaderNode ()
+ removeTailNode ()
+ moveNodeToHead (key : int)
+ moveNodeToTail (key : int)
+ clear ()
+ printList ()
+ reversePrintList
HashTable
+ table : HashNode **
+ numberOfItems : int
+ numberOfBuckets : int
+ HashTable ()
default constructor
+ HashTable (hashTableSize : int)
constructor
+ getTable () : HashNode **
+ getSize () : int
+ calculateHashCode (currentKey : int) : int
+ isEmpty () : bool
+ getNumberOfItems () : int
+ add (currKey : int , myNode : HashNode *) : bool
+ remove (currKey : int)
+ clear ()
+ getItem (currKey : int) : HashNode *
+ contains (currKey : int) : bool
+ printTable ()
Dll Node
+ statusLevel : string
+ key : int
+ fullName : string
+ address : string
+ city : string
+ state : string
+ zip : string
+ prev : DllNode *
+ next : DllNode *
+ DllNode (value : int , statusLevel : string , fullName : string , address : string
city : string , state : string , zip : string)
+ printNode ()
1 0...*
HashNode
+ key : int
+ hashCode : int
+ next : HashNode *
+ prev : HashNode *
+ DllNode * fifoNode
+ HashNode (myKey : int , myFifoNode : DllNode *)
constructor
+ getFifoNode () : DllNode
0..*
1
BinarySearchTree
- root : TreeNode *
constructor
destructor
+ BinarySearchTree () :
+ ~BinarySearchTree () :
+ addToTree (key : int)
+ removeNode (key : int) : bool
+ getHeightOfTree () : int
+ getNumberOfTreeNodes () : int
+ contains (key : int) : bool
+ getRoot () : TreeNode *
+ isEmpty () : bool
+ clear ()
+ printNodeFromTree (node : TreeNode *)
+ printInOrder ()
+ printReverseOrder ()
+ printPreOrder ()
+ printPostOrder ()
+ printDepthFirst ()
+ printBreadthFirst ()
+ printRange (low : int , high : int)
- deleteTree (node : TreeNode *)
- getHeight (node : TreeNode *)
- printInOrderHelper (node : TreeNode *)
- printReverseOrderHelper (node : TreeNode *)
- printPreOrderHelper (node : TreeNode *)
- printPostOrderHelper (node : TreeNode *)
- printRangeHelper (node : TreeNode * , low : int , high : int)
TreeNode
+ key : int
+ numberOfNodes : int
+ height : int
+ left : TreeNode *
+ right : TreeNode *
+ fifoNode : DllNode *
constructor
+ TreeNode (k : int = 0 , numNodes : int = 1 , h : int = 0 , l : TreeNode * = nullptr
r : TreeNode * = nullptr)
1
0..*

Sequence Diagram